

Chapter 4.

Number Theory and Cryptography

數論与密碼学 $\Rightarrow$ 探討整數

整數
除法

4.1 Divisibility and Modular Arithmetic

整數
表示法

4.2 Integer Representations and Algorithms

4.3 Primes and Greatest Common Divisors

4.4 Solving Congruences 重點

質數、質因數分解、GCD、最小公倍數

4.5 Applications of Congruences

4.6 Cryptography 密碼学

Chapter Motivation

Number theory (數論) is the part of mathematics devoted to the **study of the integers** and their properties. → 只探討整數和其性質

Key ideas in number theory **include divisibility and the primality of integers.**

Representations of integers, **including binary and hexadecimal** representations, are part of number theory.

Number theory has long been studied because of the beauty of its ideas, its accessibility, and its wealth of open questions.

We'll use many ideas developed in Chapter 1 about proof methods and proof strategy in our exploration of number theory.

Mathematicians have long considered number theory to be pure mathematics, but it has important applications to computer science and cryptography (密碼學) studied in Sections 4.5 and 4.6.

Divisibility and Modular Arithmetic

Section 4.1

Division

Definition: If a and b are integers with $a \neq 0$, then a divides b if there exists an integer c such that $b = ac$.

存在 a, b ; $a \neq 0$ 且 a 可被 b 整除, 則存在一 c 使得 $b = ac$

- When a divides b we say that a is a factor or divisor of b and that b is a multiple of a .

因素除數

- The notation $a \mid b$ denotes that a divides b .

- If $a \mid b$, then b/a is an integer.

- If a does not divide b , we write $a \nmid b$.

要是一整數

$$7 = 3 \times \boxed{?}$$

Example: Determine whether $3 \mid 7$ and whether

$$\boxed{3 \mid 12.}$$

✓ $12 = 3 \times \boxed{4}$

$\nmid$ $3 \nmid 7$

x

Properties of Divisibility

Theorem 1: Let a , b , and c be integers, where $a \neq 0$.

<e.g.> $a=4$

$$4 \mid 12$$

$$4 \mid 16$$

$$\therefore 4 \mid (12 \times 3 + 16 \times 5)$$

i. If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$;

ii. If $a \mid b$, then $a \mid bc$ for all integers c ; $\rightarrow \begin{matrix} b = as, & bc = asc \\ & bc = a(sc) \end{matrix} \rightarrow a \mid bc$

iii. If $a \mid b$ and $b \mid c$, then $a \mid c$.

Proof: (i) Suppose $a \mid b$ and $a \mid c$, then it follows that there are integers s and t with $b = as$ and $c = at$. Hence,

$b + c = as + at = a(s + t)$. Hence, $a \mid (b + c)$

(Exercises 3 and 4 ask for proofs of parts (ii) and (iii).)

推論 Corollary: If a , b , and c be integers, where $a \neq 0$, such that $a \mid b$ and $a \mid c$, then $a \mid mb + nc$ whenever m and n are integers. $a \mid b \therefore a \mid mb$
 $a \mid c \therefore a \mid nc$

Can you show how it follows easily from (ii) and (i) of Theorem 1?

Division Algorithm

<key> 被除數 = 除數 × 商數 + 餘數 ($a = d \times q + r$)

When an integer is divided by a positive integer, there is a quotient and a remainder. This is traditionally called the “Division Algorithm,” but is really a theorem.

Division Algorithm: If a is an integer and d a positive integer, then there are **unique** integers q and r , with $0 \leq r < d$, such that $a = dq + r$ (proved in Section 5.2).

- d is called the **divisor**. 除數
- a is called the **dividend**. 被除數
- q is called the **quotient**. 商數
- r is called the **remainder**. 餘數

Definitions of Functions
div and **mod**
 $q = \overset{101}{a} \overset{11}{\text{div}} \overset{11}{d}$
 $r = \overset{101}{a} \overset{11}{\text{mod}} \overset{11}{d}$

Examples:

- What are the quotient and remainder when 101 is divided by 11?
 $\swarrow 101 = 11 \times 9 + 2$
- Solution:** The quotient when 101 is divided by 11 is $9 = 101 \text{ div } 11$, and the remainder is $2 = 101 \text{ mod } 11$.
- What are the quotient and remainder when -11 is divided by 3?
 $\swarrow -11 = 3 \times (-4) + 1$ ✓
 ~~$-11 = 3 \times (-3) - 2$~~ → 根據定義 $r \geq 0$
- Solution:** The quotient when -11 is divided by 3 is $-4 = -11 \text{ div } 3$, and the remainder is $1 = -11 \text{ mod } 3$.

Congruence (同餘) Relation

Definition: If a and b are integers and m is a positive integer, then a is **congruent** to b modulo m if m divides $a - b$.

a當b除m時, 同餘

$$m \mid a - b$$

- The notation $a \equiv b \pmod{m}$ says that a is congruent to b modulo m .
- We say that $a \equiv b \pmod{m}$ is a *congruence* and that m is its *modulus*.
- Two integers are congruent mod m if and only if they have the same remainder when divided by m .
- If a is not congruent to b modulo m , we write $a \not\equiv b \pmod{m}$.

Example: Determine whether 17 is congruent to 5 modulo 6 and whether 24 and 14 are congruent modulo 6.

Solution: *先根據前面 divides 定義*

$$17 - 5 = 6 \times \boxed{2} \checkmark$$

- $17 \equiv 5 \pmod{6}$ because 6 divides $17 - 5 = 12$.
- $24 \not\equiv 14 \pmod{6}$ since $24 - 14 = 10$ is not divisible by 6.

$$24 - 14 = 6 \times \boxed{} \times$$

More on Congruences

Theorem 4: Let m be a positive integer. The integers a and b are congruent modulo m if and only if there is an integer k such that $a = b + km$.

Proof:

- If $a \equiv b \pmod{m}$, then (by the definition of congruence) $m \mid a - b$. Hence, there is an integer k such that $a - b = km$ and equivalently $a = b + km$.
- Conversely, if there is an integer k such that $a = b + km$, then $km = a - b$. Hence, $m \mid a - b$ and $a \equiv b \pmod{m}$.

The Relationship between (mod m) and **mod m** Notations

The use of “mod” in $a \equiv b \pmod{m}$ and $a \bmod m = b$ are different.

$$\text{let } a=22, b=18, m=4$$

$$m \mid a-b$$

$$4 \mid 22-18$$

$$22 \bmod 4 = 18 \times$$

- $a \equiv b \pmod{m}$ is a relation on the set of integers.
- In $a \bmod m = b$, the notation **mod** denotes a function.

↳ 所以这两个是不同的

The relationship between these notations is made clear in this theorem.

Theorem 3: Let a and b be integers, and let m be a positive integer. Then $a \equiv b \pmod{m}$ if and only if $a \bmod m = b \bmod m$. (Proof in the exercises)

Congruences of Sums and Products

很重要

Theorem 5: Let m be a positive integer. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$

Proof:

- Because $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, by Theorem 4 there are integers s and t with $b = a + sm$ and $d = c + tm$.
- Therefore,
 - $b + d = (a + sm) + (c + tm) = (a + c) + m(s + t)$ and
 - $bd = (a + sm)(c + tm) = ac + m(at + cs + stm)$.
- Hence, $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$.

 **Example:** Because $7 \equiv 2 \pmod{5}$ and $11 \equiv 1 \pmod{5}$, it follows from Theorem 5 that

$$18 = 7 + 11 \equiv 2 + 1 = 3 \pmod{5}$$

$$77 = 7 \cdot 11 \equiv 2 \cdot 1 = 2 \pmod{5}$$

$$\begin{array}{l} 22 \equiv 18 \pmod{4} \\ 100 \equiv 200 \pmod{4} \end{array} \Rightarrow \begin{array}{l} 22 + 100 \equiv 18 + 200 \pmod{4} \\ 22 \cdot 100 \equiv 18 \cdot 200 \pmod{4} \end{array}$$

Algebraic Manipulation of Congruences

Multiplying both sides of a valid congruence by an integer preserves validity.

$$c \equiv c \pmod{m}$$

If $a \equiv b \pmod{m}$ holds then $c \cdot a \equiv c \cdot b \pmod{m}$, where c is any integer, holds by Theorem 5 with $d = c$.

Adding an integer to both sides of a valid congruence preserves validity.

If $a \equiv b \pmod{m}$ holds then $c + a \equiv c + b \pmod{m}$, where c is any integer, holds by Theorem 5 with $d = c$.

Dividing a congruence by an integer does not always produce a valid congruence.

加、減、乘是對的，但除法不一定對！

<Link: P.56, P.57>

☆ **Example:** The congruence $14 \equiv 8 \pmod{6}$ holds. But dividing both sides by 2 does not produce a valid congruence since $14/2 = 7$ and $8/2 = 4$, but $7 \not\equiv 4 \pmod{6}$.

See Section 4.3 for conditions when division is ok.

Computing the **mod** m Function of Products and Sums

We use the following corollary to Theorem 5 to compute the remainder of the product or sum of two integers when divided by m from the remainders when each is divided by m .

推論

Corollary: Let m be a positive integer and let a and b be integers. Then $1 \cdot 2 \pmod{4} \equiv 101 \cdot 102 \pmod{4}$

$$(a + b) \pmod{m} = ((a \pmod{m}) + (b \pmod{m})) \pmod{m}$$

and

$$ab \pmod{m} = ((a \pmod{m}) (b \pmod{m})) \pmod{m}.$$

(proof in text)

Arithmetic Modulo m

Definitions: Let $\mathbb{Z}_m$ be the set of nonnegative integers less than m : $\{0, 1, \dots, m-1\}$

- The operation $+_m$ is defined as $a +_m b = (a + b) \bmod m$. This is *addition modulo m* .
- The operation $\cdot_m$ is defined as $a \cdot_m b = (a \cdot b) \bmod m$. This is *multiplication modulo m* .
- Using these operations is said to be doing *arithmetic modulo m* .

Example: Find $7 +_{11} 9$ and $7 \cdot_{11} 9$.

Solution: Using the definitions above:

- $7 +_{11} 9 = (7 + 9) \bmod 11 = 16 \bmod 11 = 5$
- $7 \cdot_{11} 9 = (7 \cdot 9) \bmod 11 = 63 \bmod 11 = 8$

Arithmetic Modulo m

The operations $+_m$ and $\cdot_m$ satisfy many of the same properties as ordinary addition and multiplication.

- Closure** (封閉性): If a and b belong to $\mathbf{Z}_m$, then $a +_m b$ and $a \cdot_m b$ belong to $\mathbf{Z}_m$.
 a, b 在作完操後，其結果是否還在集合 $\mathbf{Z}_m$
- Associativity** (結合性 (結合律)): If a, b , and c belong to $\mathbf{Z}_m$, then $(a +_m b) +_m c = a +_m (b +_m c)$ and $(a \cdot_m b) \cdot_m c = a \cdot_m (b \cdot_m c)$.
- Commutativity** (交換性 (交換率)): If a and b belong to $\mathbf{Z}_m$, then $a +_m b = b +_m a$ and $a \cdot_m b = b \cdot_m a$. $\triangle$ 矩陣乘法沒有交換率
- Identity elements** (單位元素): The elements 0 and 1 are identity elements for addition and multiplication modulo m , respectively.
 - If a belongs to $\mathbf{Z}_m$, then $a +_m 0 = a$ and $a \cdot_m 1 = a$.

加法單位元素
 $a +_m \boxed{0} = a$

乘法單位元素
 $a \cdot_m \boxed{1} = a$

Arithmetic Modulo m

反元素

- **Additive inverses:** If $a \neq 0$ belongs to $\mathbb{Z}_m$, then $m - a$ is the additive inverse of a modulo m and 0 is its own additive inverse.

- $a +_m (m - a) = 0$ and $0 +_m 0 = 0$ 乘法反元素並不總是存在!

分配

- **Distributivity:** If a , b , and c belong to $\mathbb{Z}_m$, then

- $a \cdot_m (b +_m c) = (a \cdot_m b) +_m (a \cdot_m c)$ and $(a +_m b) \cdot_m c = (a \cdot_m c) +_m (b \cdot_m c)$.

Exercises 42-44 ask for proofs of these properties.

Multiplicative inverses have not been included since they do not always exist. For example, there is no multiplicative inverse of 2 modulo 6.

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4.3 Primes and Greatest Common Divisors

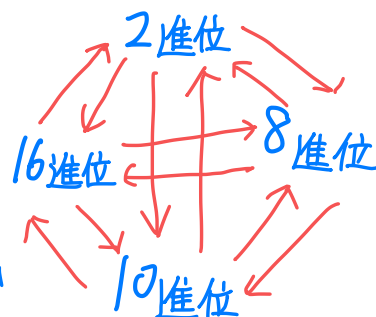
4.4 Solving Congruences

4.5 Applications of Congruences

4.6 Cryptography

4.2 重点:

寻找一个
转换进位
最快的演算法



Representations of Integers

In the modern world, we use **decimal, or base 10, notation** to represent integers. For example when we write **965**, we mean $9 \cdot 10^2 + 6 \cdot 10^1 + 5 \cdot 10^0$.

We can represent numbers using any base b , where b is a positive integer greater than 1.

The bases $b = 2$ (**binary**), $b = 8$ (**octal**), and $b = 16$ (**hexadecimal**) are important for computing and communications

The ancient Mayans used base 20 and the ancient Babylonians used base 60.

Base b Representations

We can use positive integer b greater than 1 as a base, because of this theorem:

Theorem 1: Let b be a positive integer greater than 1. Then if n is a positive integer, it can be expressed uniquely in the form:

$$n = a_k b^k + a_{k-1} b^{k-1} + \dots + a_1 b + a_0$$

where k is a nonnegative integer, $a_0, a_1, \dots, a_k$ are nonnegative integers less than b , and $a_k \neq 0$. The $a_j, j = 0, \dots, k$ are called the base- b digits of the representation.

(We will prove this using mathematical induction in Section 5.1.)

The representation of n given in Theorem 1 is called the *base b expansion of n* and is denoted by $(a_k a_{k-1} \dots a_1 a_0)_b$.

We usually omit the subscript 10 for base 10 expansions.

BIN→DEC Binary Expansions

Most computers represent integers and do arithmetic with **binary (base 2) expansions** of integers. In these expansions, the only digits used are 0 and 1.

Example: What is the decimal expansion of the integer that has $(1\ 0101\ 1111)_2$ as its binary expansion?

Solution: How to Reverse? $351 \rightarrow 10101111$ $\begin{array}{r} 175 \\ 2 \overline{) 351} \\ \underline{15} \\ 11 \\ \underline{10} \\ 1 \end{array}$ $\begin{array}{r} 87 \\ 2 \overline{) 175} \\ \underline{16} \\ 15 \\ \underline{14} \\ 1 \end{array}$ - 直除2, 直到商為0。

$$(1\ 0101\ 1111)_2 = 1 \cdot 2^8 + 0 \cdot 2^7 + 1 \cdot 2^6 + 0 \cdot 2^5 + 1 \cdot 2^4 + 1 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 = 351.$$

Example: What is the decimal expansion of the integer that has $(11011)_2$ as its binary expansion?

Solution: $(11011)_2 = 1 \cdot 2^4 + 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 = 27.$

OCT → DEC Octal Expansions

The **octal expansion (base 8)** uses the digits {0,1,2,3,4,5,6,7}.

Example: What is the decimal expansion of the number with octal expansion $(7016)_8$?

Solution: $7 \cdot 8^3 + 0 \cdot 8^2 + 1 \cdot 8^1 + 6 \cdot 8^0 = 3598$

Example: What is the decimal expansion of the number with octal expansion $(111)_8$?

Solution: $1 \cdot 8^2 + 1 \cdot 8^1 + 1 \cdot 8^0 = 64 + 8 + 1 = 73$

HEX → DEC

Hexadecimal Expansions

The **hexadecimal expansion needs 16 digits**, but our decimal system provides only 10. So letters are used for the additional symbols. The hexadecimal system uses the digits **{0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F}**. The letters **A through F** represent the decimal numbers **10 through 15**.

Example: What is the decimal expansion of the number with hexadecimal expansion **(2AE0B)₁₆**?

Solution:

$$2 \cdot 16^4 + 10 \cdot 16^3 + 14 \cdot 16^2 + 0 \cdot 16^1 + 11 \cdot 16^0 = 175627$$

Example: What is the decimal expansion of the number with hexadecimal expansion **(E5)₁₆**?

Solution: $14 \cdot 16^1 + 5 \cdot 16^0 = 224 + 5 = 229$

Base Conversion₁

Example: Find the octal expansion of $(12345)_{10}$

Solution: Successively dividing by 8 gives:

- $12345 = 8 \cdot 1543 + \mathbf{1}$

- $1543 = 8 \cdot 192 + \mathbf{7}$

- $192 = 8 \cdot 24 + \mathbf{0}$

- $24 = 8 \cdot 3 + \mathbf{0}$

- $3 = 8 \cdot 0 + \mathbf{3}$

<solver>

+進位一直除進位k,

其餘填在最後尾

一直除商,直到商為0。

The remainders are the digits from right to left yielding $\mathbf{(30071)_8}$.

Base Conversion₂

To construct the base b expansion of an integer n :

- Divide n by b to obtain a quotient and remainder.

$$n = bq_0 + \mathbf{a_0} \quad 0 \leq a_0 \leq b$$

- The remainder, a_0 , is the rightmost digit in the base b expansion of n . Next, divide q_0 by b .

$$q_0 = bq_1 + \mathbf{a_1} \quad 0 \leq a_1 \leq b$$

- The remainder, a_1 , is the second digit from the right in the base b expansion of n .
- **Continue by successively dividing the quotients by b ,** obtaining the additional base b digits as the remainder. The process terminates when the quotient is 0.

Base Conversion₃

procedure *base b expansion*(n, b : positive integers with $b > 1$)

$q := n$

$k := 0$

while ($q \neq 0$)

$a_k := q \bmod b$

$q := q \text{ div } b$

$k := k + 1$

return($a_{k-1}, \dots, a_1, a_0$) $\{(a_{k-1} \dots a_1 a_0)_b$ is base b expansion of $n\}$

q represents the quotient obtained by successive divisions by b , starting with $q = n$.

The digits in the base b expansion are the remainders of the division given by $q \bmod b$.

The algorithm terminates when $q = 0$ is reached.

Comparison of Hexadecimal, Octal, and Binary Representations

TABLE 1 Hexadecimal, Octal, and Binary Representation of the Integers 0 through 15.

Decimal	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Hexadecimal	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Octal	0	1	2	3	4	5	6	7	10	11	12	13	14	15	16	17
Binary	0	1	10	11	100	101	110	111	1000	1001	1010	1011	1100	1101	1110	1111

Initial 0s are not shown

Each octal digit corresponds to a block of 3 binary digits.

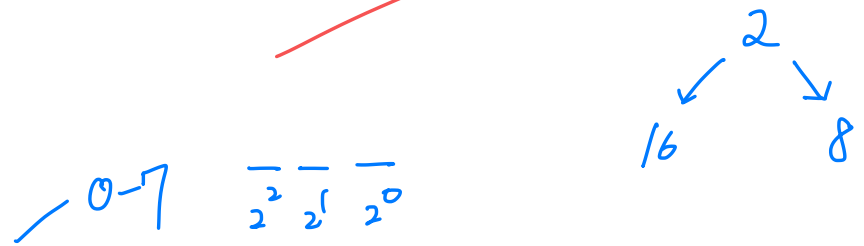
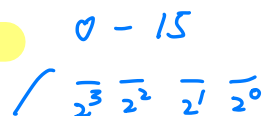
Each hexadecimal digit corresponds to a block of 4 binary digits.

So, conversion between binary, octal, and hexadecimal is easy.

Conversion Between Binary, Octal, and Hexadecimal Expansions

Example: Find the **octal** and **hexadecimal** expansions of $(11111010111100)_2$.

Solution:

- To convert to **octal**, we group the digits into **blocks of three** $(011\ 111\ 010\ 111\ 100)_2$, adding initial 0s as needed. The blocks from left to right correspond to the digits 3, 7, 2, 7, and 4. Hence, the solution is $(37274)_8$.

- To convert to **hexadecimal**, we group the digits into **blocks of four** $(0011\ 1110\ 1011\ 1100)_2$, adding initial 0s as needed. The blocks from left to right correspond to the digits 3, E, B, and C. Hence, the solution is $(3EBC)_{16}$.


Binary Addition of Integers

Algorithms for performing operations with integers using their binary expansions are important as computer chips work with **binary numbers**. Each digit is called a **bit**.

```
procedure add(a, b: positive integers)
{the binary expansions of a and b are  $(a_{n-1}, a_{n-2}, \dots, a_0)_2$  and  $(b_{n-1}, b_{n-2}, \dots, b_0)_2$ ,
respectively}
c := 0
for j := 0 to n - 1
  d :=  $\lfloor (a_j + b_j + c)/2 \rfloor$ 
  sj :=  $a_j + b_j + c - 2d$ 
  c := d
sn := c
return(s0, s1, ..., sn) {the binary expansion of the sum is  $(s_n, s_{n-1}, \dots, s_0)_2$ }
```

The number of additions of bits used by the algorithm to add two *n*-bit integers is **$O(n)$** .

Binary Addition and Subtraction

二進位
加減法

	1	0	1	0
+	1	1	1	0
1	1	0	0	0

	1	1	0	0
—	0	1	1	1
	0	1	0	1

Binary Multiplication of Integers

Algorithm for computing the product of two n bit integers.

```

procedure multiply(a, b: positive integers)
{the binary expansions of a and b are  $(a_{n-1}, a_{n-2}, \dots, a_0)_2$  and  $(b_{n-1}, b_{n-2}, \dots, b_0)_2$ , respectively}
for  $j := 0$  to  $n - 1$ 
    if  $b_j = 1$  then  $c_j = a$  shifted  $j$  places
    else  $c_j := 0$ 
{ $c_0, c_1, \dots, c_{n-1}$  are the partial products}
 $p := 0$ 
for  $j := 0$  to  $n - 1$ 
     $p := p + c_j$ 
return  $p$  { $p$  is the value of  $ab$ }
  
```

The number of additions of bits used by the algorithm to multiply two n -bit integers is $O(n^2)$.

Binary Multiplication

二進法
乘法

$$\begin{array}{r}
 101 \\
 \times 89 \\
 \hline
 909 \\
 808 \\
 \hline
 8989
 \end{array}$$

=

$$\begin{array}{r}
 01100101 \\
 \times 01011001 \\
 \hline
 01100101 \\
 01100101 \\
 01100101 \\
 01100101 \\
 \hline
 0010001100011101
 \end{array}$$

Binary Modular Exponentiation

In cryptography, it is important to be able to find $b^n \bmod m$ efficiently, where b , n , and m are large integers.

Use the binary expansion of n , $n = (a_{k-1}, \dots, a_1, a_0)_2$, to compute b^n .

Note that:

$$b^n = b^{a_{k-1} \cdot 2^{k-1} + \dots + a_1 \cdot 2 + a_0} = b^{a_{k-1} \cdot 2^{k-1}} \dots \cdot b^{a_1 \cdot 2} \cdot b^{a_0}$$

Therefore, to compute b^n , we need only compute the values of b , b^2 , $(b^2)^2 = b^4$, $(b^4)^2 = b^8$, ..., and then multiply the terms in this list, where $a_j = 1$.

Example: Compute 3^{11} using this method. *present to like BIN*

Solution: Note that $11 = (1011)_2$ so that $3^{11} = 3^8 3^2 3^1 = ((3^2)^2)^2 3^2 3^1 = (9^2)^2 \cdot 9 \cdot 3 = (81)^2 \cdot 9 \cdot 3 = 6561 \cdot 9 \cdot 3 = 177,147$.

Binary Modular Exponentiation Algorithm

Algorithm for computing the product of two n bit integers.

```
procedure modular_exponentiation( $b$ : integer,  $n = (a_{k-1}a_{k-2}\dots a_1a_0)_2$ ,  $m$ : positive integers)  
   $x := 1$   
   $power := b \bmod m$   
  for  $i := 0$  to  $k - 1$   
    if  $a_i = 1$  then  $x := (x \cdot power) \bmod m$   
     $power := (power \cdot power) \bmod m$   
  return  $x$  { $x$  equals  $b^n \bmod m$ }
```

$O((\log m)^2 \log n)$ bit operations are used to find $b^n \bmod m$.

Binary Modular Exponentiation

Find $3^{644} \bmod 645$ 典型考題!

$$644 = (1010000100)_2$$

$$3^{644} = 3^{2^9+2^7+2^2} = 3^{2^9} \cdot 3^{2^7} \cdot 3^{2^2}$$

$$3^{2^0} \bmod 645 = 3^1 \bmod 645 = 3 \bmod 645 = 3$$

$$3^{2^1} \bmod 645 = 3^2 \bmod 645 = 3^2 \bmod 645 = 9$$

$$3^{2^2} \bmod 645 = 3^4 \bmod 645 = 9^2 \bmod 645 = 81$$

$$3^{2^3} \bmod 645 = 3^8 \bmod 645 = 81^2 \bmod 645 = 111$$

$$3^{2^4} \bmod 645 = 3^{16} \bmod 645 = 111^2 \bmod 645 = 66$$

$$3^{2^5} \bmod 645 = 3^{32} \bmod 645 = 66^2 \bmod 645 = 486$$

$$3^{2^6} \bmod 645 = 3^{64} \bmod 645 = 486^2 \bmod 645 = 126$$

$$3^{2^7} \bmod 645 = 3^{128} \bmod 645 = 126^2 \bmod 645 = 396$$

$$3^{2^8} \bmod 645 = 3^{2^{56}} \bmod 645 = 396^{2^2} \bmod 645 = 81$$

$$3^{2^9} \bmod 645 = 3^{512} \bmod 645 = 81^2 \bmod 645 = 111$$

拆三項相乘

此效率其實對於電腦很快
但對於人算非常沒效率 ✓

4.4 會介紹更好的 way

Link 费用小定理, P.70

$$3^{644} \pmod{3}$$

$$3^{644} \pmod{6}$$

$3^{64} \pmod{43}$

$$3^{644} \bmod 645 = (3^{2^9} \cdot 3^{2^7} \cdot 3^{2^2}) \bmod 645 = (81 \cdot 396 \cdot 111) \bmod 645 = 36$$

Chapter 4.

Number Theory and Cryptography

4.1 Divisibility and Modular Arithmetic

4.2 Integer Representations and Algorithms

4.3 Primes and Greatest Common Divisors

4.4 Solving Congruences

4.5 Applications of Congruences

4.6 Cryptography

Primes

— 正整數!

大於 1

Definition: A positive integer p greater than 1 is called **prime** (質數) if the only positive factors of p are 1 and p . A positive integer that is greater than 1 and is not prime is called **composite**.
合數 / 非質數

Example: The integer 7 is prime because its only positive factors are 1 and 7, but 9 is composite because it is divisible by 3.

The Fundamental Theorem of

質因數分解

Arithmetic

Theorem: Every positive integer greater than 1 can be written uniquely as a prime or as the product of two or more primes where the prime factors are written in order of nondecreasing size.

Examples:

- $100 = 2 \cdot 2 \cdot 5 \cdot 5 = 2^2 \cdot 5^2$
- $641 = 641$
- $999 = 3 \cdot 3 \cdot 3 \cdot 37 = 3^3 \cdot 37$
- $1024 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 2^{10}$

證明質數

Infinitude of Primes

無限多(反證法)

Theorem: There are infinitely many primes.

Proof: Assume finitely many primes: $p_1, p_2, \dots, p_n$ 都>1! 假设質數有限

- Let $q = p_1 p_2 \cdots p_n + 1 \rightarrow$ let $q =$ 所有質數相乘+1 也是質數 与假设矛盾!
- Either q is prime or by the fundamental theorem of arithmetic it is a product of primes.
 - But none of the primes p_j divides q since if $p_j \mid q$, then p_j divides $q - p_1 p_2 \cdots p_n = 1$.
 - Hence, there is a prime not on the list $p_1, p_2, \dots, p_n$. It is either q , or if q is composite, it is a prime factor of q . This contradicts the assumption that $p_1, p_2, \dots, p_n$ are all the primes.
- Consequently, there are infinitely many primes.

Representing Functions



Marin Mersenne

(1588-1648)

✓ p 与 $2^p - 1$ 之結果都要是 Prime

Definition: **Prime numbers** of the form **$2^p - 1$** , where **p is prime**, are called **Mersenne primes (梅森質數)**.

- $2^2 - 1 = 3$, $2^3 - 1 = 7$, $2^5 - 1 = 37$, and $2^7 - 1 = 127$ are Mersenne primes.
- $2^{11} - 1 = 2047$ is not a Mersenne prime since $2047 = 23 \cdot 89$.

Distribution of Primes

Mathematicians have been interested in the distribution of prime numbers among the positive integers. In the nineteenth century, the *prime number theorem* was proved which gives an asymptotic estimate for the number of primes not exceeding x .

$$\begin{aligned}\pi(10) &= 4 & (2, 3, 5, 7) \\ \pi(11) &= 5 & (2, 3, 5, 7, 11)\end{aligned}$$

Prime Number Theorem: The ratio of $\pi(x)$, the number of primes not exceeding x , and $x/\ln x$ approaches 1 as x grows without bound. ($\ln x$ is the natural logarithm of x)

- The theorem tells us that the number of primes not exceeding x , can be approximated by $x/\ln x$.
- The odds that a randomly selected positive integer less than n is prime are approximately $(n/\ln n)/n = 1/\ln n$.

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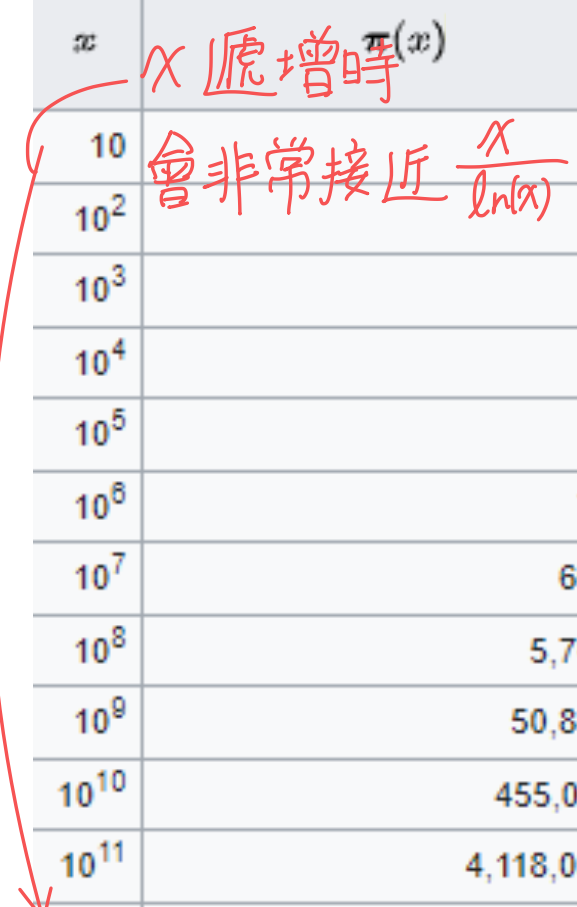
Prime Number Theorem: The ratio of $\pi(x)$, the number of primes not exceeding x , and $x / \ln x$ approaches 1 as x grows without bound. ($\ln x$ is the natural logarithm of x)

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隨機選擇到的機率約為 $1 / \ln n$

- The odds that a randomly selected positive integer less than n is prime are approximately $(n / \ln n) / n = 1 / \ln n$.

Prime Number Theorem: The ratio of $\pi(x)$, the number of primes not exceeding x , and $x/\ln x$ approaches 1 as x grows without bound. ($\ln x$ is the natural logarithm of x)



x 遞增時
會非常接近 $\frac{x}{\ln(x)}$

x	$\pi(x)$	$\frac{\pi(x)}{x/\ln x}$
10	4	0.921
10^2	25	1.151
10^3	168	1.161
10^4	1,229	1.132
10^5	9,592	1.104
10^6	78,498	1.084
10^7	664,579	1.071
10^8	5,761,455	1.061
10^9	50,847,534	1.054
10^{10}	455,052,511	1.048
10^{11}	4,118,054,813	1.043
10^{12}	37,607,912,018	1.039

10^{13}	346,065,536,839	1.034
10^{14}	3,204,941,750,802	1.033
10^{15}	29,844,570,422,669	1.031
10^{16}	279,238,341,033,925	1.029
10^{17}	2,623,557,157,654,233	1.027
10^{18}	24,739,954,287,740,860	1.025
10^{19}	234,057,667,276,344,607	1.024
10^{20}	2,220,819,602,560,918,840	1.023
10^{21}	21,127,269,486,018,731,928	1.022
10^{22}	201,467,286,689,315,906,290	1.021
10^{23}	1,925,320,391,606,803,968,923	1.020
10^{24}	18,435,599,767,349,200,867,866	1.019
10^{25}	176,846,309,399,143,769,411,680	1.018

Generating Primes

The problem of generating large primes is of both theoretical and practical interest.

We will see (in Section 4.6) that **finding large primes** with hundreds of digits **is important in cryptography**.

So far, no useful closed formula that always produces primes has been found. There is no simple function $f(n)$ such that $f(n)$ is prime for all positive integers n .

But $f(n) = n^2 - n + 41$ is prime for all integers $1, 2, \dots, 40$. Because of this, we might conjecture (推測) that $f(n)$ is prime for all positive integers n . But $f(41) = 41^2$ is not prime.

不存在一个多项式

More generally, **there is no polynomial with integer coefficients such that $f(n)$ is prime for all positive integers n** . (See supplementary Exercise 23.)

Conjectures about Primes

Even though primes have been studied extensively for centuries, many conjectures about them are unresolved, including:

Goldbach's Conjecture (哥德巴赫猜想): Every even (偶數) integer n , $n > 2$, is the sum of two primes. It has been verified by computer for all positive even integers up to $1.6 \cdot 10^{18}$. The conjecture is believed to be true by most mathematicians.

The Twin Prime Conjecture (孪生質數猜想): The twin prime conjecture is that there are infinitely many pairs of twin primes. Twin primes are pairs of primes that differ by 2. Examples are 3 and 5, 5 and 7, 11 and 13, etc. The current world's record for twin primes (as of mid 2011) consists of numbers $65,516,468,355 \cdot 23^{33,333} \pm 1$, which have 100,355 decimal digits.

Greatest Common Divisor₁

GCD 最大公因數

Definition: Let a and b be integers, **not both zero**. The largest integer d such that $d \mid a$ and also $d \mid b$ is called the **greatest common divisor (最大公因數)** of a and b . The greatest common divisor of a and b is denoted by **$\gcd(a,b)$** .

One can find greatest common divisors of small numbers by inspection. *How to calculate GCD?* ① 短除法 ② 質因數分解 ③ 輾轉相除法

e.g. $\gcd(24, 36)$

Example: What is the greatest common divisor of 24 and 36?

Solution: $\gcd(24, 36) = 12$

Example: What is the greatest common divisor of 17 and 22?

Solution: $\gcd(17, 22) = 1$

①

2	24	36
	12	18
2	6	9
	3	3

12

②

$24 = 2^3 \times 3$
 $36 = 2^2 \times 3^2$
 $\rightarrow 2^2 \times 3 = 12$

③

2	24	36	1
	24	24	
	0	12	

Greatest Common Divisor₂

Definition: The integers a and b are **relatively prime (互質)** if their **greatest common divisor is 1**.

Example: 17 and 22

Definition: The integers $a_1, a_2, \dots, a_n$ are **pairwise relatively prime (兩兩互質)** if **$\gcd(a_i, a_j) = 1$ whenever $1 \leq i < j \leq n$** .

Example: Determine whether the integers 10, 17 and 21 are pairwise relatively prime.

Solution: Because $\gcd(10, 17) = 1$, $\gcd(10, 21) = 1$, and $\gcd(17, 21) = 1$, 10, 17, and 21 are **pairwise relatively prime**.

Example: Determine whether the integers 10, 19, and 24 are pairwise relatively prime.

Solution: Because $\gcd(10, 24) = 2$, 10, 19, and 24 are not pairwise relatively prime.

46 Finding the Greatest Common Divisor Using Prime Factorizations

Suppose the prime factorizations of a and b are:

$$a = p_1^{a_1} p_2^{a_2} \cdots p_n^{a_n}, \quad b = p_1^{b_1} p_2^{b_2} \cdots p_n^{b_n},$$

where each exponent is a nonnegative integer, and where all primes occurring in either prime factorization are included in both. Then:

$$\gcd(a, b) = p_1^{\min(a_1, b_1)} p_2^{\min(a_2, b_2)} \cdots p_n^{\min(a_n, b_n)},$$

This formula is valid since the integer on the right (of the equals sign) divides both a and b . No larger integer can divide both a and b .

Example: $120 = 2^3 \cdot 3 \cdot 5$ $500 = 2^2 \cdot 5^3$ $\Rightarrow 2^2 \times 5 = 20$

$$\gcd(120, 500) = 2^{\min(3, 2)} \cdot 3^{\min(1, 0)} \cdot 5^{\min(1, 3)} = 2^2 \cdot 3^0 \cdot 5^1 = 20$$

Finding the gcd of two positive integers using their prime factorizations is not efficient because there is no efficient algorithm for finding the prime factorization of a positive integer.

Least Common Multiple

Definition: The **least common multiple (最小公倍數)** of the positive integers a and b is the smallest positive integer that is divisible by both a and b . It is denoted by $\text{lcm}(a, b)$.

The least common multiple can also be computed from the prime factorizations.

$$\text{lcm}(a, b) = p_1^{\max(a_1, b_1)} p_2^{\max(a_2, b_2)} \cdots p_n^{\max(a_n, b_n)},$$

This number is divided by both a and b and no smaller number is divided by a and b .

Example: $\text{lcm}(2^3 3^5 7^2, 2^4 3^3) = 2^{\max(3, 4)} 3^{\max(5, 3)} 7^{\max(2, 0)} = 2^4 3^5 7^2$

The greatest common divisor and the least common multiple of two integers are related by:

Theorem 5: Let a and b be positive integers. Then

$$ab = \text{gcd}(a, b) \cdot \text{lcm}(a, b)$$

(proof is Exercise 31)

$$\text{let } a=20, b=15$$

$$ab=300, \text{gcd}(20, 15)=5, \text{lcm}(20, 15)=60$$

Euclidean Algorithm₁

The **Euclidian algorithm** (歐幾里得算法，輾轉相除法) is an efficient method for computing the greatest common divisor of two integers. It is based on the idea that $\gcd(a,b)$ is equal to $\gcd(a,c)$ when $a > b$ and c is the remainder when a is divided by b .

Example: Find $\gcd(91, 287)$:

- $287 = 91 \cdot 3 + 14$
(7x41) (7x13)
- $91 = 14 \cdot 6 + 7$
(7x13) (7x2)
- $14 = 7 \cdot 2 + 0$
(7x2) (7x1)

Stopping condition

Divide 287 by 91

Divide 91 by 14

Divide 14 by 7

$$\begin{array}{r|l} 91 & 287 \\ \hline 84 & 3 \\ \hline 7 & 14 \\ \hline 0 & 0 \end{array}$$

GCD



Euclid

(325 B.C.E. – 265 B.C.E.)

$$\gcd(287, 91) = \gcd(91, 14) = \gcd(14, 7) = 7$$

Euclidean Algorithm₂

The Euclidean algorithm expressed in pseudocode is:

```
procedure gcd(a, b: positive integers)
```

```
x := a
```

```
y := b
```

```
while y ≠ 0
```

```
    r := x mod y
```

```
    x := y
```

```
    y := r
```

```
return x {gcd(a,b) is x}
```

In Section 5.3, we'll see that the time complexity of the algorithm is $O(\log b)$, where $a > b$.

Correctness of Euclidean Algorithm₁

Lemma 1: Let $a = bq + r$, where a, b, q , and r are integers. Then $\gcd(a, b) = \gcd(b, r)$.

Proof:

- Suppose that d divides both a and b . Then d also divides $a - bq = r$ (by Theorem 1 of Section 4.1). Hence, any common divisor of a and b must also be any common divisor of b and r .
- Suppose that d divides both b and r . Then d also divides $bq + r = a$. Hence, any common divisor of a and b must also be a common divisor of b and r .
- Therefore, $\gcd(a, b) = \gcd(b, r)$.

Correctness of Euclidean Algorithm₂

Suppose that a and b are positive integers with $a \geq b$.

Let $r_0 = a$ and $r_1 = b$.

Successive applications of the division algorithm yields:

$$\begin{array}{ll} r_0 = r_1 q_1 + r_2 & 0 \leq r_2 < r_1 \\ r_1 = r_2 q_2 + r_3 & 0 \leq r_3 < r_2 \\ \vdots & \\ r_{n-2} = r_{n-1} q_{n-1} + r_n & 0 \leq r_n < r_{n-1} \\ r_{n-1} = r_n q_n & \end{array}$$

199	=	2	×	72	+	55
72	=	1	×	55	+	17
55	=	3	×	17	+	4
17	=	4	×	4	+	1
4	=	4	×	1	+	0
GCD(199, 72) = 1						

Eventually, a remainder of zero occurs in the sequence of terms: $a = r_0 > r_1 > r_2 > \cdots \geq 0$. The sequence can't contain more than a terms.

By Lemma 1

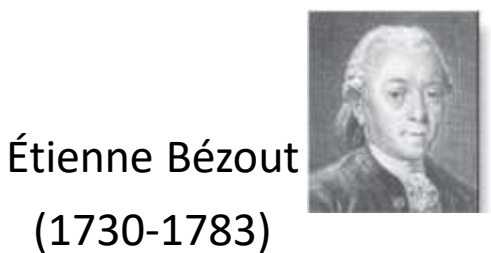
$$\gcd(a, b) = \gcd(r_0, r_1) = \cdots = \gcd(r_{n-1}, r_n) = \gcd(r_n, 0) = r_n.$$

Hence the greatest common divisor is the last nonzero remainder in the sequence of divisions.

这泥馬

非常重要 > <

gcds as Linear Combinations



Bézout's Theorem (貝祖定理): If a and b are positive integers, then there exist integers s and t such that $\gcd(a,b) = sa + tb$. 存在 $\gcd(a,b) = sa + tb$ 的線性組合!

(proof in exercises of Section 5.2) $5 = 20s + 15t$, 整數 s, t 是否存在? Yes!
 $43a + 17b = 1$, 整數 a, b 是否存在? Yes!

Definition: If a and b are positive integers, then integers s and t such that $\gcd(a,b) = sa + tb$ are called **Bézout coefficients** of a and b . The equation $\gcd(a,b) = sa + tb$ is called **Bézout's identity (貝祖等式)**.

By Bézout's Theorem, the gcd of integers a and b can be expressed in the form $sa + tb$ where s and t are integers. **This is a linear combination with integer coefficients of a and b .**

- $\gcd(6,14) = (-2) \cdot 6 + 1 \cdot 14$

Graphs of Functions

$$252a + 198b = 18, a \cdot b \text{ Existed!}$$

Example: Express $\gcd(252, 198) = 18$ as a linear combination of 252 and 198.

Solution: First use the Euclidean algorithm to show $\gcd(252, 198) = 18$

i. $252 = 1 \cdot 198 + 54$

ii. $198 = 3 \cdot 54 + 36$

iii. $54 = 1 \cdot 36 + 18$

iv. $36 = 2 \cdot 18$

- Now working backwards, from iii and i above

- $18 = 54 - 1 \cdot 36$

- $36 = 198 - 3 \cdot 54$

- Substituting the 2nd equation into the 1st yields:

- $18 = 54 - 1 \cdot (198 - 3 \cdot 54) = 4 \cdot 54 - 1 \cdot 198$

- Substituting $54 = 252 - 1 \cdot 198$ (from i)) yields:

- $18 = 4 \cdot (252 - 1 \cdot 198) - 1 \cdot 198 = 4 \cdot 252 - 5 \cdot 198$

This method illustrated above is a two pass method. It first uses the Euclidian algorithm to find the gcd and then works backwards to express the gcd as a linear combination of the original two integers. A one pass method, called the *extended Euclidean algorithm*, is developed in the exercises.

Consequences of Bézout's Theorem

Lemma 2: If a, b , and c are positive integers such that $\gcd(a, b) = 1$ and $a \mid bc$, then $a \mid c$.

let $a=5, b=4$

$\rightarrow 5 \mid (4 \times 10), \quad 5 \mid 10$

Proof: Assume $\gcd(a, b) = 1$ and $a \mid bc$

- Since $\gcd(a, b) = 1$, by Bézout's Theorem there are integers s and t such that $sa + tb = 1$.
- Multiplying both sides of the equation by c , yields $sac + tbc = c$.
- From Theorem 1 of Section 4.1:
 $a \mid tbc$ (part ii) and a divides $sac + tbc$ since $a \mid sac$ and $a \mid tbc$ (part i)
- We conclude $a \mid c$, since $sac + tbc = c$.

- CH5 數學歸納法

Lemma 3: If p is prime and $p \mid a_1 a_2 \cdots a_n$, then $p \mid a_i$ for some i .

(proof uses mathematical induction; see Exercise 64 of Section 5.1)

Lemma 3 is crucial in the proof of the uniqueness of prime factorizations.

Uniqueness of Prime Factorization

Conclusion: 質因數分解是唯一的

We will prove that a prime factorization of a positive integer where **the primes are in nondecreasing order is unique**. (This part of the fundamental theorem of arithmetic. The other part, which asserts that every positive integer has a prime factorization into primes, will be proved in Section 5.2.)

Proof: (*by contradiction*) Suppose that the positive integer n can be written as a product of primes in two distinct ways:

$$n = p_1 p_2 \cdots p_s \text{ and } n = q_1 q_2 \cdots q_t.$$

- **Remove all common primes** from the factorizations to get $p_{i_1} p_{i_2} \cdots p_u = q_{j_1} q_{j_2} \cdots q_{j_v}$
- By Lemma 3, **it follows that p_{i_1} divides q_{j_k} , for some k** , contradicting the assumption that and are distinct primes.
- Hence, there can be at most one factorization of n into primes in nondecreasing order.

Dividing Congruences by an Integer₁

Dividing both sides of a valid congruence by an integer does not always produce a valid congruence (see Section 4.1).

11

Algebraic Manipulation of Congruences

Multiplying both sides of a valid congruence by an integer preserves validity.

If $a \equiv b \pmod{m}$ holds then $c \cdot a \equiv c \cdot b \pmod{m}$, where c is any integer, holds by Theorem 5 with $d = c$.

Adding an integer to both sides of a valid congruence preserves validity.

If $a \equiv b \pmod{m}$ holds then $c + a \equiv c + b \pmod{m}$, where c is any integer, holds by Theorem 5 with $d = c$.

Dividing a congruence by an integer does not always produce a valid congruence.

Example: The congruence $14 \equiv 8 \pmod{6}$ holds. But dividing both sides by 2 does not produce a valid congruence since $14/2 = 7$ and $8/2 = 4$, but $7 \not\equiv 4 \pmod{6}$.

See Section 4.3 for conditions when division is ok.

Dividing Congruences by an Integer₂

兩個、或兩個以上整數的gcd為1，
則稱它們為互質

But dividing by an integer relatively prime (互質) to the modulus does produce a valid congruence:

當兩數互質時，
除法是有效的

dividing by 2

32%7 = 4 18%7 = 4 32 ≡ 18 (mod 7)

16%7 = 2 9%7 = 4 16 ≡ 9 (mod 7)

Theorem 7: Let m be a positive integer and let a , b , and c be integers. If $ac \equiv bc \pmod{m}$ and $\gcd(c, m) = 1$, then $a \equiv b \pmod{m}$.

Proof: Since $ac \equiv bc \pmod{m}$, $m \mid ac - bc = c(a - b)$ by Lemma 2 and the fact that $\gcd(c, m) = 1$, it follows that $m \mid a - b$. Hence, $a \equiv b \pmod{m}$.

Chapter 4.

Number Theory and Cryptography

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Linear Congruences

三大類問題：

- ①: $3x \equiv 4 \pmod{5}$
- ②: $x \equiv 1 \pmod{3}$
 $x \equiv 2 \pmod{5}$
 $x \equiv 1 \pmod{7}$
- ③: $3x \equiv 4 \pmod{5}$
 $2x \equiv 6 \pmod{7}$
 $5x \equiv 1 \pmod{11}$

Definition: A congruence of the form

$ax \equiv b \pmod{m},$

where m is a positive integer, a and b are integers, and x is a variable, is called a **linear congruence (一次同餘方程式)**.

The solutions to a linear congruence $ax \equiv b \pmod{m}$ are all integers x that satisfy the congruence.

Definition: An integer $\bar{a}$ such that $\bar{a}a \equiv 1 \pmod{m}$ is said to be an **inverse (反元素)** of a modulo m .

① Solution 1: 用湊的, Since mod 7, $x=0 \sim 6$ 湊

② Next Page: 貝祖定理

Example: 5 is an inverse of 3 modulo 7 since $5 \cdot 3 = 15 \equiv 1 \pmod{7}$

乘法反元素

One method of solving linear congruences makes use of an inverse $\bar{a}$, if it exists. Although we can not divide both sides of the congruence by a , we can multiply by $\bar{a}$ to solve for x .

Inverse of a modulo m

The following theorem guarantees that an inverse of a modulo m exists whenever a and m are relatively prime. Two integers a and b are relatively prime when $\gcd(a,b) = 1$.

Theorem 1: If a and m are relatively prime integers and $m > 1$, then an inverse of a modulo m exists. Furthermore, this inverse is unique modulo m . (This means that there is a unique positive integer $\bar{a}$ less than m that is an inverse of a modulo m and every other inverse of a modulo m is congruent to $\bar{a}$ modulo m .)

Proof: Since $\gcd(a,m) = 1$, by Theorem 6 of Section 4.3, there are integers s and t such that $sa + tm = 1$.

兩邊同除 m !

- Hence, $sa + tm \equiv 1 \pmod{m}$.
- Since $tm \equiv 0 \pmod{m}$, it follows that $sa \equiv 1 \pmod{m}$.
- Consequently, s is an inverse of a modulo m .
- The uniqueness of the inverse is Exercise 7.

Finding Inverses₁

The Euclidean algorithm and Bézout coefficients gives us a systematic approaches to finding inverses.

Example: Find an inverse of 3 modulo 7.

Solution: Because $\gcd(3,7) = 1$, by Theorem 1, an inverse of 3 modulo 7 exists.

- Using the Euclidian algorithm: $7 = 2 \cdot 3 + 1$.
- From this equation, we get $-2 \cdot 3 + 1 \cdot 7 = 1$, and see that -2 and 1 are Bézout coefficients of 3 and 7.
- Hence, -2 is an inverse of 3 modulo 7.
- Also every integer congruent to -2 modulo 7 is an inverse of 3 modulo 7, i.e., 5, -9, 12, etc.

Finding Inverses₂

$\Rightarrow 10|x \equiv 1 \pmod{4620}$

Example: Find an inverse of 101 modulo 4620.

Solution: First use the Euclidian algorithm to show that $\gcd(101,4620) = 1$.

$4620 = 45 \cdot 101 + 75$

$101 = 1 \cdot 75 + 26$

$75 = 2 \cdot 26 + 23$

$26 = 1 \cdot 23 + 3$

$23 = 7 \cdot 3 + 2$

$3 = 1 \cdot 2 + 1$

$2 = 2 \cdot 1$

Since the last nonzero remainder is 1, $\gcd(101,4260) = 1$

Working Backwards:

$1 = 3 - 1 \cdot 2$

$1 = 3 - 1 \cdot (23 - 7 \cdot 3) = -1 \cdot 23 + 8 \cdot 3$

$1 = -1 \cdot 23 + 8 \cdot (26 - 1 \cdot 23) = 8 \cdot 26 - 9 \cdot 23$

$1 = 8 \cdot 26 - 9 \cdot (75 - 2 \cdot 26) = 26 \cdot 26 - 9 \cdot 75$

$1 = 26 \cdot (101 - 1 \cdot 75) - 9 \cdot 75$
 $= 26 \cdot 101 - 35 \cdot 75$

$1 = 26 \cdot 101 - 35 \cdot (4620 - 45 \cdot 101)$
 $= -35 \cdot 4620 + 1601 \cdot 101$

歐幾里得
算法

建構
貝祖等式

Bézout coefficients : - 35 and 1601

1601 is an inverse of 101 modulo 4620

Using Inverses to Solve Congruences

We can solve the congruence $ax \equiv b \pmod{m}$ by multiplying both sides by $\bar{a}$.

$$x \equiv 5 \times 3x \equiv 5 \times 4 \equiv 6 \pmod{7}$$

Example: What are the solutions of the congruence $3x \equiv 4 \pmod{7}$.

Solution: We found that -2 is an inverse of 3 modulo 7 (two slides back). We multiply both sides of the congruence by -2 giving

$$-2 \cdot 3x \equiv -2 \cdot 4 \pmod{7}.$$

Because $-6 \equiv 1 \pmod{7}$ and $-8 \equiv 6 \pmod{7}$, it follows that if x is a solution, then $x \equiv -8 \equiv 6 \pmod{7}$

We need to determine if every x with $x \equiv 6 \pmod{7}$ is a solution. Assume that $x \equiv 6 \pmod{7}$. By Theorem 5 of Section 4.1, it follows that $3x \equiv 3 \cdot 6 = 18 \equiv 4 \pmod{7}$ which shows that all such x satisfy the congruence.

The solutions are the integers x such that $x \equiv 6 \pmod{7}$, namely, $6, 13, 20, \dots$ and $-1, -8, -15, \dots$

The Chinese Remainder Theorem₁

中国剩餘定理

In the first century, the Chinese mathematician Sun-Tsu asked:

There are certain things whose number is unknown. When divided by 3, the remainder is 2; when divided by 5, the remainder is 3; when divided by 7, the remainder is 2. What will be the number of things?

This puzzle can be translated into the solution of the system of congruences:

$x \equiv 2 \pmod{3},$

$x \equiv 3 \pmod{5},$

$x \equiv 2 \pmod{7}?$

<key 还原! $p \times q + r$ > 0, 1, 2, 3, 4, ...

$x = 3 \times q + 2 = 3 \times (5r + s) + 2 = 15r + 3s + 2$

$15r + 8 = 15 \times (7a + b) + 8$ 0, 1, 2, 3, 4, 5, 6

$= 105a + 15b + 8$ try 0, 1, 2, ... 6

when $\div 5$, 餘 2 $\Rightarrow 105a + 23$, Ans = 23

5 try 0, 1, 2, 3, 4
when $\div 5$, 餘 3
 $\hookrightarrow s = 2$
 $15r + 8$

We'll see how the theorem that is known as the *Chinese Remainder Theorem* can be used to solve Sun-Tsu's problem.

$$x \equiv 2 \pmod{4}$$

$$x \equiv 1 \pmod{6}$$

$$x \equiv 2 \pmod{7}$$

find x .

$$\Rightarrow x = 4q + 2$$

0, 1, 2, 3, 4, 5

$$\begin{aligned} \textcircled{1} &= 4(6r + \textcircled{s}) + 2 \\ &= 24r + 4s + 2 \end{aligned}$$

$$\begin{aligned} \checkmark \textcircled{2} &= 4(3r + \textcircled{s}) + 2 \\ &= 12r + 4s + 2 \end{aligned}$$

0, 1, 2

有LCM的话

优先选小的!

設 $x \equiv 2 \pmod{3}$, $x \equiv 3 \pmod{5}$, $x \equiv 2 \pmod{7}$, 求 $x = ?$ ◀

$$\text{由題意可知} \begin{cases} 3|x-2 \cdots \cdots \textcircled{1} \\ 5|x-3 \cdots \cdots \textcircled{2} \\ 7|x-2 \cdots \cdots \textcircled{3} \end{cases}$$

設 $x = 3 \times 5 \times 7t + 3 \times 5a + 5 \times 7b + 3 \times 7c$, 代入①中得 $3|35b-2$, 取 $b=1$, ◀

x 代入②中得 $5|21c-3$, 取 $c=3$, ◀

x 代入③中得 $7|15a-2$, 取 $a=2$, ◀

所以 $x = 105t + 30 + 35 + 63 = 105t + 128 = 105(t+1) + 23$, 故 x 的最小值為 23 。 ◀

The Chinese Remainder Theorem₂

Theorem 2: (*The Chinese Remainder Theorem*) Let $m_1, m_2, \dots, m_n$ be pairwise relatively prime positive integers greater than one and $a_1, a_2, \dots, a_n$ arbitrary integers. Then the system

$$x \equiv a_1 \pmod{m_1}$$

$$x \equiv a_2 \pmod{m_2}$$

.

.

.

$$x \equiv a_n \pmod{m_n}$$

has a unique solution modulo $m = m_1 m_2 \cdots m_n$.

(That is, there is a solution x with $0 \leq x < m$ and all other solutions are congruent modulo m to this solution.)

Proof: We'll show that a solution exists by describing a way to construct the solution. Showing that the solution is unique modulo m is Exercise 30.

The Chinese Remainder Theorem₃

To construct a solution first let $M_k = m/m_k$ for $k = 1, 2, \dots, n$ and $m = m_1 m_2 \cdots m_n$. Since $\gcd(m_k, M_k) = 1$, by Theorem 1, there is an integer y_k , an inverse of M_k modulo m_k , such that

$$M_k y_k \equiv 1 \pmod{m_k}.$$

Form the sum

$$x = a_1 M_1 y_1 + a_2 M_2 y_2 + \cdots + a_n M_n y_n.$$

Note that because $M_j \equiv 0 \pmod{m_k}$ whenever $j \neq k$, all terms except the k th term in this sum are congruent to 0 modulo m_k .

Because $M_k y_k \equiv 1 \pmod{m_k}$, we see that $x \equiv a_k M_k y_k \equiv a_k \pmod{m_k}$, for $k = 1, 2, \dots, n$.

Hence, x is a simultaneous solution to the n congruences.

$$x \equiv a_1 \pmod{m_1}$$

$$x \equiv a_2 \pmod{m_2}$$

.

.

.

$$x \equiv a_n \pmod{m_n}$$

The Chinese Remainder Theorem₄

Example: Consider the 3 congruences from Sun-Tsu's problem:

$$x \equiv 2 \pmod{3}, \quad x \equiv 3 \pmod{5}, \quad x \equiv 2 \pmod{7}.$$

- Let $m = 3 \cdot 5 \cdot 7 = 105$, $M_1 = m/3 = 35$, $M_2 = m/5 = 21$, $M_3 = m/7 = 15$.
- We see that
 - 2 is an inverse of $M_1 = 35$ modulo 3 since $35 \cdot 2 \equiv 2 \cdot 2 \equiv 1 \pmod{3}$
 - 1 is an inverse of $M_2 = 21$ modulo 5 since $21 \equiv 1 \pmod{5}$
 - 1 is an inverse of $M_3 = 15$ modulo 7 since $15 \equiv 1 \pmod{7}$
- Hence,

$$x = a_1 M_1 y_1 + a_2 M_2 y_2 + a_3 M_3 y_3$$

$$= 2 \cdot 35 \cdot 2 + 3 \cdot 21 \cdot 1 + 2 \cdot 15 \cdot 1 = 233 \equiv 23 \pmod{105}$$
- We have shown that 23 is the smallest positive integer that is a simultaneous solution. Check it!



Back Substitution

同後替代法

We can also solve systems of linear congruences with pairwise relatively prime moduli by rewriting a congruence as an equality using Theorem 4 in Section 4.1, substituting the value for the variable into another congruence, and continuing the process until we have worked through all the congruences. This method is known as *back substitution*.

Example: Use the method of back substitution to find all integers x such that $x \equiv 1 \pmod{5}$, $x \equiv 2 \pmod{6}$, and $x \equiv 3 \pmod{7}$.

Solution: By Theorem 4 in Section 4.1, the first congruence can be rewritten as $x = 5t + 1$, where t is an integer.

- Substituting into the second congruence yields $5t + 1 \equiv 2 \pmod{6}$.
- Solving this tells us that $t \equiv 5 \pmod{6}$.
- Using Theorem 4 again gives $t = 6u + 5$ where u is an integer.
- Substituting this back into $x = 5t + 1$, gives $x = 5(6u + 5) + 1 = 30u + 26$.
- Inserting this into the third equation gives $30u + 26 \equiv 3 \pmod{7}$.
- Solving this congruence tells us that $u \equiv 6 \pmod{7}$.
- By Theorem 4, $u = 7v + 6$, where v is an integer.
- Substituting this expression for u into $x = 30u + 26$, tells us that $x = 30(7v + 6) + 26 = 210v + 206$.

Translating this back into a congruence we find the solution $x \equiv 206 \pmod{210}$.

$$3x \equiv 4 \pmod{5} \rightarrow \boxed{2} 3x = \boxed{2} 4 \pmod{5}$$

$$2x \equiv 7 \pmod{11}$$

$$5x \equiv 1 \pmod{13}$$

$$\begin{array}{l} \text{|||} \\ 6x \\ \text{|||} \\ x \end{array} \quad \begin{array}{l} \text{|||} \\ 3 \\ \swarrow \end{array} \quad x \equiv 3 \pmod{5}$$

$$\xrightarrow{\text{还原}} x \equiv 9 \pmod{11}$$

$$x \equiv 8 \pmod{13}$$

The Solution
like Case 2.

Fermat's Little Theorem

費馬小定理

Theorem 3: (Fermat's Little Theorem) If p is prime and a is an integer not divisible by p , then

$a^{p-1} \equiv 1 \pmod{p}$

Furthermore, for every integer a we have

$a^p \equiv a \pmod{p}$

但 p 必為
質數

注意，若 p 不是質數則此式不成立！

反例： $p=4, a=3$
 $3^{4-1} \equiv 3 \not\equiv 1 \pmod{4}$

(proof outlined in Exercise 19)

Fermat's little theorem is useful in computing the remainders modulo p of large powers of integers.

Example: Find $7^{222} \pmod{11}$.

By Fermat's little theorem, we know that $7^{10} \equiv 1 \pmod{11}$, and so $(7^{10})^k \equiv 1 \pmod{11}$, for every positive integer k . Therefore,

$7^{222} = 7^{22 \cdot 10 + 2} = (7^{10})^{22} 7^2 \equiv (1)^{22} \cdot 49 \equiv 5 \pmod{11}$.

Hence, $7^{222} \pmod{11} = 5$.



Pierre de Fermat (1601-1665)

Pseudoprimes₁

By Fermat's little theorem $n > 2$ is prime, where

$$2^{n-1} \equiv 1 \pmod{n}.$$

But if this congruence holds, n may not be prime. **Composite** integers n such that $2^{n-1} \equiv 1 \pmod{n}$ are called **pseudoprimes (偽質數)** to the base 2.

Example: The integer **341 is a pseudoprime to the base 2.**

$$341 = 11 \cdot 31$$

成立!，但他不是質數!

$$2^{340} \equiv 1 \pmod{341} \text{ (see in Exercise 37)}$$

We can replace 2 by any integer $b \geq 2$.

Definition: Let b be a positive integer. If n is a composite integer, and $b^{n-1} \equiv 1 \pmod{n}$, then n is called a **pseudoprime to the base b .**

Pseudoprimes₂

Given a positive integer n , such that $2^{n-1} \equiv 1 \pmod{n}$:

- If n does not satisfy the congruence, it is **composite**. 合數
- If n does satisfy the congruence, it is **either prime or a pseudoprime to the base 2**. 質數 偽質數

Doing similar tests with additional bases b , provides more evidence as to whether n is prime.

Among the positive integers not exceeding a positive real number x , compared to primes, there are **relatively few pseudoprimes to the base b** .

- For example, among the positive integers less than 10^{10} there are 455,052,512 primes, but only 14,884 pseudoprimes to the base 2.

Primitive Roots

Definition: A **primitive root (原根)** modulo a prime p is an integer r in $\mathbf{Z}_p$ such that every nonzero element of $\mathbf{Z}_p$ is a power of r .

Example: Since every element of $\mathbf{Z}_{11}$ is a power of 2, 2 is a primitive root of 11.
 $\mathbf{Z}_{11} = \{0, 1, 2, \dots, 10\}$

Powers of 2 modulo 11: $2^1 = 2, 2^2 = 4, 2^3 = 8, 2^4 = 5, 2^5 = 10, 2^6 = 9, 2^7 = 7, 2^8 = 3, 2^9 = 6, 2^{10} = 1$.

Example: Since not all elements of $\mathbf{Z}_{11}$ are powers of 3, 3 is not a primitive root of 11.

Powers of 3 modulo 11: $3^1 = 3, 3^2 = 9, 3^3 = 5, 3^4 = 4, 3^5 = 1$, and the pattern repeats for higher powers.

Important Fact: There is a primitive root modulo p for every prime number p .

Discrete Logarithms

Suppose p is prime and r is a primitive root modulo p . If a is an integer between 1 and $p - 1$, that is an element of $\mathbb{Z}_p$, there is a unique exponent e such that $r^e = a$ in $\mathbb{Z}_p$, that is, $r^e \bmod p = a$.

Definition: Suppose that p is prime, r is a primitive root modulo p , and a is an integer between 1 and $p - 1$. If $r^e \bmod p = a$ and $1 \leq e \leq p - 1$, we say that e is the **discrete logarithm** (離散對數) of a modulo p to the base r and we write $\log_r a = e$ (where the prime p is understood).

Example 1: We write $\log_2 3 = 8$ since the discrete logarithm of 3 modulo 11 to the base 2 is 8 as $2^8 = 3 \bmod 11$.

Example 2: We write $\log_2 5 = 4$ since the discrete logarithm of 5 modulo 11 to the base 2 is 4 as $2^4 = 5 \bmod 11$.

There is no known polynomial time algorithm for computing the discrete logarithm of a modulo p to the base r (when given the prime p , a root r modulo p , and a positive integer $a \in \mathbb{Z}_p$). The problem plays a role in cryptography as will be discussed in Section 4.6.

Chapter 4.

Number Theory and Cryptography

4.1 Divisibility and Modular Arithmetic

4.2 Integer Representations and Algorithms

4.3 Primes and Greatest Common Divisors

4.4 Solving Congruences

4.5 Applications of Congruences

4.6 Cryptography

Hashing Functions

Definition: A **hashing function** (雜湊函數) h assigns memory location $h(k)$ to the record that has k as its key.

- A common hashing function is $h(k) = k \bmod m$, where m is the number of memory locations.
- Because **this hashing function is onto**, all memory locations are possible.

Example: Let $h(k) = k \bmod 111$. This hashing function assigns the records of customers with social security numbers as keys to memory locations in the following manner:

$$h(064212848) = 064212848 \bmod 111 = 14$$

$$h(037149212) = 037149212 \bmod 111 = 65$$

$$h(107405723) = 107405723 \bmod 111 = 14, \text{ but } \text{since location 14 is already occupied, the record is assigned to the next available position, which is 15.}$$

The hashing function is not one-to-one as there are many more possible keys than memory locations. When more than one record is assigned to the same location, we say a *collision* occurs. Here a collision has been resolved by assigning the record to the first free location.

For collision resolution, we can use a *linear probing function*:

$$h(k, i) = (h(k) + i) \bmod m, \text{ where } i \text{ runs from } 0 \text{ to } m - 1.$$

There are many other methods of handling with collisions. You may cover these in a later CS course.

Pseudorandom Numbers₁

Randomly chosen numbers are needed for many purposes, including computer simulations.

Pseudorandom numbers are **not truly random** since they are generated by systematic methods.

The linear congruential method is one commonly used procedure for generating pseudorandom numbers.

Four integers are needed: the *modulus* m , the *multiplier* a , the *increment* c , and *seed* x_0 , with $2 \leq a < m$, $0 \leq c < m$, $0 \leq x_0 < m$.

We generate a sequence of pseudorandom numbers $\{x_n\}$, with $0 \leq x_n < m$ for all n , by successively using the recursively defined function

$$x_{n+1} = (ax_n + c) \bmod m.$$

(an example of a recursive definition, discussed in Section 5.3)

If psuedorandom numbers between 0 and 1 are needed, then the generated numbers are **divided by the modulus, x_n/m .**

Pseudorandom Numbers₂

Example: Find the sequence of pseudorandom numbers generated by the linear congruential method with modulus $m = 9$, multiplier $a = 7$, increment $c = 4$, and seed $x_0 = 3$.

Solution: Compute the terms of the sequence by successively using the congruence

$$x_{n+1} = (7x_n + 4) \bmod 9, \text{ with } x_0 = 3$$

$$x_1 = (7x_0 + 4) \bmod 9 = (7 \cdot 3 + 4) \bmod 9 = 25 \bmod 9 = 7$$

$$x_2 = (7x_1 + 4) \bmod 9 = (7 \cdot 7 + 4) \bmod 9 = 53 \bmod 9 = 8$$

$$x_3 = (7x_2 + 4) \bmod 9 = (7 \cdot 8 + 4) \bmod 9 = 60 \bmod 9 = 6$$

$$x_4 = (7x_3 + 4) \bmod 9 = (7 \cdot 6 + 4) \bmod 9 = 46 \bmod 9 = 1$$

$$x_5 = (7x_4 + 4) \bmod 9 = (7 \cdot 1 + 4) \bmod 9 = 11 \bmod 9 = 2$$

$$x_6 = (7x_5 + 4) \bmod 9 = (7 \cdot 2 + 4) \bmod 9 = 18 \bmod 9 = 0$$

$$x_7 = (7x_6 + 4) \bmod 9 = (7 \cdot 0 + 4) \bmod 9 = 4 \bmod 9 = 4$$

$$x_8 = (7x_7 + 4) \bmod 9 = (7 \cdot 4 + 4) \bmod 9 = 32 \bmod 9 = 5$$

$$x_9 = (7x_8 + 4) \bmod 9 = (7 \cdot 5 + 4) \bmod 9 = 39 \bmod 9 = 3$$

The sequence generated is 3,7,8,6,1,2,0,4,5,3,7,8,6,1,2,0,4,5,3,...

It repeats after generating 9 terms.

Commonly, computers use a linear congruential generator with **increment $c = 0$. This is called a *pure multiplicative generator*.** Such a generator with **modulus $2^{31} - 1$** and **multiplier $7^5 = 16,807$** generates $2^{31} - 2$ numbers before repeating.

Check Digits: UPCs

A common method of detecting errors in strings of digits is to add an extra digit at the end, which is evaluated using a function. If the final digit is not correct, then the string is assumed not to be correct.

Example: Retail products are identified by their *Universal Product Codes (UPCs)* (通用產品代碼). Usually these have 12 decimal digits, the last one being the check digit. The check digit is determined by the congruence:

$$3x_1 + x_2 + 3x_3 + x_4 + 3x_5 + x_6 + 3x_7 + x_8 + 3x_9 + x_{10} + 3x_{11} + x_{12} \equiv 0 \pmod{10}.$$

- Suppose that the first 11 digits of the UPC are **79357343104**. What is the check digit?
- Is **041331021641** a valid UPC?

Solution:

- $$3 \cdot 7 + 9 + 3 \cdot 3 + 5 + 3 \cdot 7 + 3 + 3 \cdot 4 + 3 + 3 \cdot 1 + 0 + 3 \cdot 4 + x_{12} \equiv 0 \pmod{10}$$

$$21 + 9 + 9 + 5 + 21 + 3 + 12 + 3 + 3 + 0 + 12 + x_{12} \equiv 0 \pmod{10}$$

$$\mathbf{98 + x_{12} \equiv 0 \pmod{10}}$$

$$x_{12} \equiv 2 \pmod{10} \text{ So, the check digit is 2.}$$
- $$3 \cdot 0 + 4 + 3 \cdot 1 + 3 + 3 \cdot 3 + 1 + 3 \cdot 0 + 2 + 3 \cdot 1 + 6 + 3 \cdot 4 + 1 \equiv 0 \pmod{10}$$

$$\mathbf{0 + 4 + 3 + 3 + 9 + 1 + 0 + 2 + 3 + 6 + 12 + 1 = 44 \equiv 4 \not\equiv 0 \pmod{10}}$$

Hence, **041331021641** is not a valid UPC.

Check Digits: ISBNs

Books are identified by an *International Standard Book Number (ISBN-10)* (國際標準書號), a 10 digit code. The first 9 digits identify the language, the publisher, and the book. The tenth digit is a check digit, which is determined by the following congruence

$$x_{10} \equiv \sum_{i=1}^9 ix_i \pmod{11}.$$

$$\sum_{i=1}^{10} ix_i \equiv 0 \pmod{11}.$$

The validity of an ISBN-10 number can be evaluated with the equivalent

- Suppose that the first 9 digits of the ISBN-10 are **007288008**. What is the check digit?
- Is **084930149X** a valid ISBN10?

因為除以11可能會得到10這個餘數 ISBN會用X這個符號來表示10。

Solution:

$$a. \quad X_{10} \equiv 1 \cdot 0 + 2 \cdot 0 + 3 \cdot 7 + 4 \cdot 2 + 5 \cdot 8 + 6 \cdot 8 + 7 \cdot 0 + 8 \cdot 0 + 9 \cdot 8 \pmod{11}.$$

$$X_{10} \equiv 0 + 0 + 21 + 8 + 40 + 48 + 0 + 0 + 72 \pmod{11}.$$

$$X_{10} \equiv 189 \equiv 2 \pmod{11}. \text{ Hence, } X_{10} = 2.$$

$$b. \quad 1 \cdot 0 + 2 \cdot 8 + 3 \cdot 4 + 4 \cdot 9 + 5 \cdot 3 + 6 \cdot 0 + 7 \cdot 1 + 8 \cdot 4 + 9 \cdot 9 + 10 \cdot 10 =$$

$$0 + 16 + 12 + 36 + 15 + 0 + 7 + 32 + 81 + 100 = 299 \equiv 2 \not\equiv 0 \pmod{11}$$

Hence, **084930149X** is not a valid ISBN-10.

A *single error* is an error in **one digit** of an identification number and a *transposition error* is the accidental **interchanging of two digits**. Both of these kinds of errors can be detected by the check digit for ISBN-10. (see text for more details)

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4.6 Cryptography

Caesar Cipher₁

(凱撒密碼)



Julius Caesar created secret messages by **shifting each letter three letters** forward in the alphabet (sending the last three letters to the first three letters.) For example, the letter B is replaced by E and the letter X is replaced by A. This process of making a message secret is an example of **encryption** (加密).

Here is how the encryption process works:

- Replace each letter by an integer from $\mathbb{Z}_{26}$, that is an integer from 0 to 25 representing one less than its position in the alphabet.
- The encryption function is **$f(p) = (p + 3) \bmod 26$** . It replaces each integer p in the set $\{0, 1, 2, \dots, 25\}$ by $f(p)$ in the set $\{0, 1, 2, \dots, 25\}$.
- Replace each integer p by the letter with the position $p + 1$ in the alphabet.

Example: Encrypt the message “**MEET YOU IN THE PARK**” using the Caesar cipher.

Solution: 12 4 4 19 24 14 20 8 13 19 7 4 15 0 17 10.

Now replace each of these numbers p by **$f(p) = (p + 3) \bmod 26$** .

15 7 7 22 1 17 23 11 16 22 10 7 18 3 20 13.

Translating the numbers back to letters produces the encrypted message

“**PHHW BRX LQ WKH SDUN.**”

Caesar Cipher₂

To **recover the original message**, use $f^{-1}(p) = (p-3) \bmod 26$. So, each letter in the coded message is shifted back three letters in the alphabet, with the first three letters sent to the last three letters. This process of recovering the original message from the encrypted message is called **decryption** (解密).

The Caesar cipher is one of a family of ciphers called **shift ciphers**. Letters can be shifted by an integer k , with 3 being just one possibility. The encryption function is

$$f(p) = (p + k) \bmod 26$$

and the decryption function is

$$f^{-1}(p) = (p-k) \bmod 26$$

The integer k is called a **key**.

Shift Cipher₁

Example 1: Encrypt the message “STOP GLOBAL WARMING” using the shift cipher with $k = 11$.

Solution: Replace each letter with the corresponding element of $\mathbb{Z}_{26}$.

18 19 14 15 6 11 14 1 0 11 22 0 17 12 8 13 6.

Apply the shift $f(p) = (p + 11) \bmod 26$, yielding

3 4 25 0 17 22 25 12 11 22 7 11 2 23 19 24 17.

Translating the numbers back to letters produces the ciphertext

“DEZA RWZMLW HLCXTYR.”

Shift Cipher₂

Example 2: Decrypt the message “**LEWLYPLUJL PZ H NYLHA ALHJOLY**” that was encrypted using the shift cipher with $k = 7$.

Solution: Replace each letter with the corresponding element of $\mathbb{Z}_{26}$.

11 4 22 11 24 15 11 20 9 1 15 25 7 13 24 11 7 0 0 11 7 9 14
11 24.

Shift each of the numbers by $-k = -7$ modulo 26, yielding

4 23 15 4 17 8 4 13 2 4 8 18 0 6 17 4 0 19 19 4 0 2 7 4 17.

Translating the numbers back to letters produces the decrypted message

“**EXPERIENCE IS A GREAT TEACHER.**”

Affine Ciphers

(仿射密碼)

Shift ciphers are a special case of **affine ciphers** which use functions of the form

$$f(p) = (ap + b) \bmod 26,$$

where a and b are integers, chosen so that f is a bijection.

The function is a bijection if and only if $\gcd(a, 26) = 1$.

Example: What letter replaces the letter K when the function $f(p) = (7p + 3) \bmod 26$ is used for encryption.

Solution: Since 10 represents K, $f(10) = (7 \cdot 10 + 3) \bmod 26 = 21$, which is then replaced by V.

To decrypt a message encrypted by a shift cipher, the congruence $c \equiv ap + b \pmod{26}$ needs to be solved for p .

- **Subtract b** from both sides to obtain $c - b \equiv ap \pmod{26}$.
- **Multiply both sides by the inverse** of a modulo 26, which exists since $\gcd(a, 26) = 1$.
- $\bar{a}(c - b) \equiv \bar{a}ap \pmod{26}$, which simplifies to $\bar{a}(c - b) \equiv p \pmod{26}$.
- $p \equiv \bar{a}(c - b) \pmod{26}$ is used to determine p in $\mathbf{Z}_{26}$.

Cryptanalysis of Affine Ciphers

The process of recovering plaintext from ciphertext **without knowledge both of the encryption method and the key** is known as **cryptanalysis** (密碼分析) or **breaking codes** (密碼破解).

An important tool for cryptanalyzing ciphertext produced with a affine ciphers is the relative frequencies of letters. **The nine most common letters in the English texts are E 13%, T 9%, A 8%, O 8%, I 7%, N 7%, S 7%, H 6%, and R 6%.**

To analyze ciphertext:

- Find the frequency of the letters in the ciphertext.
- Hypothesize that the most frequent letter is produced by encrypting E.
- If the value of the shift from E to the most frequent letter is k , shift the ciphertext by $-k$ and see if it makes sense.
- If not, try T as a hypothesis and continue.

Example: We intercepted the message “**ZNK KGXRE HOXJ MKZY ZNK CUXS**” that we know was produced by a shift cipher. Let’s try to cryptanalyze.

Solution: **The most common letter in the ciphertext is K. So perhaps the letters were shifted by 6 since this would then map E to K.** Shifting the entire message by -6 gives us “**THE EARLY BIRD GETS THE WORM.**”

Block Ciphers₁

(區塊密碼)

Ciphers that replace each letter of the alphabet by another letter are called *character* or *monoalphabetic* ciphers.

They are vulnerable to cryptanalysis based on letter frequency. **Block ciphers** avoid this problem, by replacing blocks of letters with other blocks of letters.

A simple type of block cipher is called the **transposition cipher**. The key is a permutation σ of the set $\{1, 2, \dots, m\}$, where m is an integer, that is a one-to-one function from $\{1, 2, \dots, m\}$ to itself.

To encrypt a message, split the letters into blocks of size m , adding additional letters to fill out the final block. We encrypt $p_1, p_2, \dots, p_m$ as $c_1, c_2, \dots, c_m = p_{\sigma(1)}, p_{\sigma(2)}, \dots, p_{\sigma(m)}$.

To decrypt the $c_1, c_2, \dots, c_m$ transpose the letters using the inverse permutation σ^{-1} .

Block Ciphers₂

Example: Using the transposition cipher based on the permutation σ of the set $\{1,2,3,4\}$ with $\sigma(1) = 3$, $\sigma(2) = 1$, $\sigma(3) = 4$, $\sigma(4) = 2$,

- Encrypt the plaintext **PIRATE ATTACK**
- Decrypt the ciphertext message **SWUE TRAEOEHS**, which was encrypted using the same cipher.

Solution:

- Split into four blocks** PIRA TEAT TACK.
Apply the permutation σ giving **IAPR ETTA AKTC**.
- σ^{-1} : $\sigma^{-1}(1) = 2$, $\sigma^{-1}(2) = 4$, $\sigma^{-1}(3) = 1$, $\sigma^{-1}(4) = 3$.

Apply the permutation σ^{-1} giving USEW ATER HOSE.

Split into words to obtain USE WATER HOSE.

Cryptosystems₁

Definition: A **cryptosystem** (密碼系統) is a five-tuple (P, C, K, E, D) , where

- P is the set of plaintext strings,
- C is the set of ciphertext strings,
- K is the **keyspace** (set of all possible keys),
- E is the set of encryption functions, and
- D is the set of decryption functions.

The encryption function in E corresponding to the key k is denoted by E_k and the decryption function in D that decrypts cipher text encrypted using E_k is denoted by D_k . Therefore:

$$D_k(E_k(p)) = p, \text{ for all plaintext strings } p.$$

Cryptosystems₂

Example: Describe the family of shift ciphers as a cryptosystem.

Solution: Assume the messages are strings consisting of elements in $\mathbf{Z}_{26}$.

- P is the set of strings of elements in $\mathbf{Z}_{26}$,
- C is the set of strings of elements in $\mathbf{Z}_{26}$,
- $K = \mathbf{Z}_{26}$,
- E consists of functions of the form $E_k(p) = (p + k) \bmod 26$, and
- D is the same as E where $D_k(p) = (p - k) \bmod 26$.